Concept 3 | Rational Expressions

English Student Edition • Focused formulas and guided practice

Formula opener

Restriction

$$\text{denominator} \neq 0$$

Multiply / divide

$$\frac{A}{B} \times \frac{C}{D} = \frac{AC}{BD}$$

Add / subtract

$$\frac{A}{B} \pm \frac{C}{D} = \frac{AD \pm BC}{BD}$$

Eng Eslam Ahmed | 01120009622

Classified Practice **Rational Expressions**

Q001 **Challenge**

After simplifying this transfer chain before substitution, which result is correct for $2\,()$?:

Choose the correct option.

A $\frac{2x(x+3)}{(x-3)(x-1)}$

B $\frac{-2x^2-6x}{x^2-4x+3}$

C $\frac{4x^2+12x}{x^2-4x+3}$

D $\frac{x^2+3x}{x^2-4x+3}$

Classified Practice **Rational Expressions**

Q002 Challenge

After simplifying this transfer chain before substitution, which result is correct for 3 ()?:

Choose the correct option.

A $\frac{-3x^2+3x+18}{x^2+3x}$

B $\frac{3(x-3)(x+2)}{x(x+3)}$

C $\frac{6x^2-6x-36}{x^2+3x}$

D $\frac{3x^2-3x-18}{2x^2+6x}$

Classified Practice **Rational Expressions**

Q003 Challenge

After simplifying this transfer chain before substitution, which result is correct for $4\,()$?:

Choose the correct option.

A $\frac{-8x^2+24x+32}{x^2-5x+6}$

B $\frac{16x^2-48x-64}{x^2-5x+6}$

C $\frac{8(x-4)(x+1)}{(x-3)(x-2)}$

D $\frac{4x^2-12x-16}{x^2-5x+6}$

Classified Practice **Rational Expressions**

Q004 **Challenge**

After simplifying this transfer chain before substitution, which result is correct for $8\,()$?:

Choose the correct option.

A $-\frac{8}{x}$

B $\frac{16}{x}$

C $\frac{4}{x}$

D $\frac{8}{x}$

Classified Practice **Rational Expressions**

Q005 **Challenge**

After simplifying this transfer chain before substitution, which result is correct for $9\,()$**?:**

Choose the correct option.

A $\frac{9(x-1)(x^4+x^3+x^2+x+1)}{2}$

B $-\frac{9x^5}{2}+\frac{9}{2}$

C $9x^5-9$

D $\frac{9x^5}{4}-\frac{9}{4}$

Classified Practice **Rational Expressions**

Q006 Challenge

After simplifying this transfer chain before substitution, which result is correct for $10\,()$?:

Choose the correct option.

A $\frac{-10x^3+10x^2+160x-160}{x^3-5x^2+6x}$

B $\frac{10(x-4)(x-1)(x+4)}{x(x-3)(x-2)}$

C $\frac{20x^3-20x^2-320x+320}{x^3-5x^2+6x}$

D $\frac{5x^3-5x^2-80x+80}{x^3-5x^2+6x}$

Classified Practice **Rational Expressions**

Q007 **Written**

Simplify $\frac{12ab^5}{9a^3b^2}$.

Q008 **Written**

Simplify $\frac{6x^2y^3}{20x^3y}$.

Classified Practice **Rational Expressions**

Q009 **Written**

Simplify $\frac{6a^2bc^2}{8ab^4c^2}$.

Q010 **Written**

Simplify $\frac{2a^2b}{8a^3b^3}$.

Classified Practice **Rational Expressions**

Q011 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{12ab^5}{9a^3b^2}\right)$?:

Choose the correct option.

A $-\frac{4b^3}{a^2}$

B $\frac{4b^3}{a^2}$

C $\frac{8b^3}{a^2}$

D $\frac{2b^3}{a^2}$

Q012 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{6x^2y^3}{20x^3y}\right)$?:

Choose the correct option.

A $\frac{9y^2}{10x}$

B $-\frac{9y^2}{10x}$

C $\frac{9y^2}{5x}$

D $\frac{9y^2}{20x}$

Classified Practice **Rational Expressions**

Q013 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{6a^2bc^2}{8ab^4c^2}\right)$?:

Choose the correct option.

A $\frac{3a}{b^3}$

B $-\frac{3a}{b^3}$

C $\frac{6a}{b^3}$

D $\frac{3a}{2b^3}$

Q014 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{2a^2b}{8a^3b^3}\right)$?:

Choose the correct option.

A $-\frac{1}{2ab^2}$

B $\frac{1}{2ab^2}$

C $\frac{1}{ab^2}$

D $\frac{1}{4ab^2}$

Classified Practice **Rational Expressions**

Q015 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{18a^3b^2}{24ab^5}\right)$?:

Choose the correct option.

A $\frac{3a^2}{2b^3}$

B $-\frac{3a^2}{2b^3}$

C $\frac{3a^2}{b^3}$

D $\frac{3a^2}{4b^3}$

Q016 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{-15x^4y^3}{20x^2y}\right)$?:

Choose the correct option.

A $\frac{9x^2y^2}{4}$

B $-\frac{9x^2y^2}{4}$

C $-\frac{9x^2y^2}{2}$

D $-\frac{9x^2y^2}{8}$

Classified Practice **Rational Expressions**

Q017 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{45a^2b^4c}{60a^5bc^2}\right)$?:

Choose the correct option.

A $-\frac{3b^3}{a^3c}$

B $\frac{6b^3}{a^3c}$

C $\frac{3b^3}{a^3c}$

D $\frac{3b^3}{2a^3c}$

Q018 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{14x^5y^2}{21x^2y^4}\right)$?:

Choose the correct option.

A $-\frac{4x^3}{3y^2}$

B $\frac{8x^3}{3y^2}$

C $\frac{2x^3}{3y^2}$

D $\frac{4x^3}{3y^2}$

Classified Practice **Rational Expressions**

Q019 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{32a^4}{48ab^2}\right)$?:

Choose the correct option.

A $\frac{2a^3}{b^2}$

B $-\frac{2a^3}{b^2}$

C $\frac{4a^3}{b^2}$

D $\frac{a^3}{b^2}$

Q020 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{27x^3y^5}{36xy^2}\right)$?:

Choose the correct option.

A $-3x^2y^3$

B $3x^2y^3$

C $6x^2y^3$

D $\frac{3x^2y^3}{2}$

Classified Practice **Rational Expressions**

Q021 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-42a^2c^3}{56a^5c}\right)$?:

Choose the correct option.

A $\frac{3c^2}{2a^3}$

B $-\frac{3c^2}{a^3}$

C $-\frac{3c^2}{2a^3}$

D $-\frac{3c^2}{4a^3}$

Q022 **Written**

Simplify $\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}$.

Classified Practice **Rational Expressions**

Q023 **Written**

Simplify $\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}$.

Q024 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{a^3b^2}{3c} \times \frac{9c^3}{a^4b^2}\right)$?:

Choose the correct option.

A $-\frac{12c^2}{a}$

B $\frac{24c^2}{a}$

C $\frac{12c^2}{a}$

D $\frac{6c^2}{a}$

Classified Practice **Rational Expressions**

Q025 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{6bc^3}{5a^2} \times \frac{10a^4}{3b^3}\right)$?:

Choose the correct option.

A $-\frac{8a^2c^3}{b^2}$

B $\frac{16a^2c^3}{b^2}$

C $\frac{4a^2c^3}{b^2}$

D $\frac{8a^2c^3}{b^2}$

Q026 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{6a^2b}{5c} \times \frac{15c^2}{4ab^3}\right)$?:

Choose the correct option.

A $-\frac{27ac}{2b^2}$

B $\frac{27ac}{b^2}$

C $\frac{27ac}{4b^2}$

D $\frac{27ac}{2b^2}$

Classified Practice **Rational Expressions**

Q027 MCQ

After reducing coefficients and powers, which result is correct for $4\left(\frac{9x^3y}{14a^2} \times \frac{21a^3}{6xy^2}\right)$?:

Choose the correct option.

A $\frac{9ax^2}{y}$

B $-\frac{9ax^2}{y}$

C $\frac{18ax^2}{y}$

D $\frac{9ax^2}{2y}$

Q028 MCQ

After reducing coefficients and powers, which result is correct for $2\left(\frac{-8ab^2}{3c^2} \times \frac{12c^3}{-5a^2b}\right)$?:

Choose the correct option.

A $-\frac{64bc}{5a}$

B $\frac{64bc}{5a}$

C $\frac{128bc}{5a}$

D $\frac{32bc}{5a}$

Classified Practice **Rational Expressions**

Q029 MCQ

After reducing coefficients and powers, which result is correct for $3\left(\frac{10x^2y^3}{9ab} \times \frac{27a^2b^2}{25xy}\right)$?:

Choose the correct option.

A $-\frac{18abxy^2}{5}$

B $\frac{36abxy^2}{5}$

C $\frac{18abxy^2}{5}$

D $\frac{9abxy^2}{5}$

Q030 **Written**

Simplify $\frac{x^2-4}{3x^2+7x+2}$.

Classified Practice **Rational Expressions**

Q031 **Written**

Simplify $\frac{2x^2-3x-5}{x^2+3x+2}$.

Q032 **Written**

Simplify $\frac{x^2+3x}{x^2-9}$.

Classified Practice **Rational Expressions**

Q033 **Written**

Simplify $\frac{x^2+x-2}{x^2-5x+4}$.

Q034 **Written**

Simplify $\frac{2x^2-4xy+2y^2}{x^3+3x^2y-4xy^2}$.

Classified Practice **Rational Expressions**

Q035 **Written**

Simplify $\frac{x^3-1}{x^2+2x-3}$.

Q036 **Written**

Simplify $\frac{x^2-11x+24}{x^2-6x-16} \times \frac{x^2+2x}{x^2-6x+9}$.

Classified Practice **Rational Expressions**

Q037 **Written**

Simplify $\frac{x+1}{x^2-4} \times \frac{x^2-2x-8}{x^2+x}$.

Q038 **Written**

Simplify $\frac{x^2-9}{x^2-2x} \times \frac{x^2-7x+10}{x^2+x-12}$.

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Q039 **Written**

Simplify $\frac{x^2+x-6}{2x^2+x-6} \times \frac{x^2+6x+8}{3x^2-5x-2}$.

Q040 **Written**

Simplify $\frac{x^2-4x-5}{2x^2+5x+2} \div \frac{x^2-x-20}{2x^2-5x-3}$.

Classified Practice **Rational Expressions**

Q041 **Written**

Simplify $\frac{x^2-9}{x^2-x-2} \div \frac{2x^2+5x-3}{x+1}$.

Q042 **Written**

Simplify $\frac{2xy}{3ab} \div \frac{8x^3y}{9a^2b^3}$.

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Q043 **Written**

Simplify $\frac{2x^2-x-10}{x^2-2x-3} \div \frac{x^2+x-2}{x^2-6x+9}$.

Q044 **Written**

Simplify $\frac{x^2-3x-10}{2x^2+3x-2} \div \frac{x^2-2x-15}{6x-3}$.

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Q045 **Written**

Simplify $\frac{x^2-y^2}{x^2-2xy+y^2} \div \frac{x^2+xy}{x-y}$.

Q046 **Written**

Simplify $\frac{x^3+8}{x^2-x-6} \times \frac{x^2-6x+9}{x^2-2x+4} \div \frac{2x-6}{x^5-1}$ **for** $x = \sqrt{123}$**.**

Classified Practice **Rational Expressions**

Q047 Challenge

Simplify
$\left(\frac{x+1}{x-2}-\frac{2}{x+3}+\frac{3}{x+3}-\frac{1}{x-2}\right)$; **state every excluded value.**

Classified Practice **Rational Expressions**

Q048 **Written**

Simplify

$$\frac{x^2-4}{x+1}+\frac{3}{x+1}.$$

Classified Practice **Rational Expressions**

Q049 MCQ

Simplify

$$\frac{x^2+2x+3}{x+1}+\frac{2x-1}{x+1}:$$

Choose the correct option.

A $\frac{x^2+5x+3}{x+1}$

B $\frac{x^2+3x+1}{x+1}$

C $\frac{x^2+4x+2}{x+1}$

D $-\frac{x^2+4x+2}{x+1}$

Classified Practice **Rational Expressions**

Q050 MCQ

Simplify

$$\frac{2x^2-5}{x^2-4}+\frac{x+6}{x^2-4}:$$

Choose the correct option.

A $\frac{3x^2+x-3}{(x-2)(x+2)}$

B $\frac{x^2+x+5}{(x-2)(x+2)}$

C $-\frac{2x^2+x+1}{(x-2)(x+2)}$

D $\frac{2x^2+x+1}{(x-2)(x+2)}$

Classified Practice **Rational Expressions**

Q051 MCQ

Simplify

$$\frac{3x^2+x-2}{(x-2)(x+1)}+\frac{4x-3}{(x-2)(x+1)}:$$

Choose the correct option.

A $\frac{3x^2+5x-5}{(x-2)(x+1)}$

B $\frac{4x^2+4x-7}{(x-2)(x+1)}$

C $\frac{2x^2+6x-3}{(x-2)(x+1)}$

D $-\frac{3x^2+5x-5}{(x-2)(x+1)}$

Classified Practice **Rational Expressions**

Q052 MCQ

Simplify

$$\frac{x^3-1}{x+2}+\frac{2x^2+x+4}{x+2}:$$

Choose the correct option.

A $\frac{x^3+2x^2+2x+5}{x+2}$

B $\frac{x^3+2x^2+x+3}{x+2}$

C $\frac{x^3+2x^2+1}{x+2}$

D $-\frac{x^3+2x^2+x+3}{x+2}$

Classified Practice **Rational Expressions**

Q053 **Written**

Simplify

$$\frac{3}{x^2+x}-\frac{-x+3}{x^2+x}.$$

Classified Practice **Rational Expressions**

Q054 MCQ

Simplify

$\dfrac{x^2+4x+4}{x+2}-\dfrac{x+4}{x+2}$:

Choose the correct option.

A $\dfrac{x(x+3)}{x+2}$

B $\dfrac{x^2+4x+2}{x+2}$

C $\dfrac{x^2+2x-2}{x+2}$

D $-\dfrac{x(x+3)}{x+2}$

Classified Practice **Rational Expressions**

Q055 MCQ

Simplify

$$\frac{3x^2-2x+1}{x^2-1}-\frac{2x-5}{x^2-1}:$$

Choose the correct option.

A $\dfrac{4x^2-4x+5}{(x-1)(x+1)}$

B $\dfrac{3x^2-4x+6}{(x-1)(x+1)}$

C $\dfrac{2x^2-4x+7}{(x-1)(x+1)}$

D $-\dfrac{3x^2-4x+6}{(x-1)(x+1)}$

Classified Practice **Rational Expressions**

Q056 MCQ

Simplify

$$\frac{x^2-6x+8}{(x-1)(x+2)}-\frac{x^2-3x+2}{(x-1)(x+2)}:$$

Choose the correct option.

A $\frac{x^2-2x+4}{(x-1)(x+2)}$

B $-\frac{x^2+4x-8}{(x-1)(x+2)}$

C $-\frac{3(x-2)}{(x-1)(x+2)}$

D $\frac{3(x-2)}{(x-1)(x+2)}$

Classified Practice **Rational Expressions**

Q057 MCQ

Simplify

$$\frac{2x^3 + x}{x - 3} - \frac{x^2 + 7}{x - 3}:$$

Choose the correct option.

A $\dfrac{2x^3 - x^2 + 2x - 10}{x - 3}$

B $\dfrac{2x^3 - x^2 - 4}{x - 3}$

C $\dfrac{2x^3 - x^2 + x - 7}{x - 3}$

D $-\dfrac{2x^3 - x^2 + x - 7}{x - 3}$

Classified Practice **Rational Expressions**

Q058 **Written**

Simplify

$$\frac{x+4}{x^2-2x}-\frac{3}{x^2-3x+2}.$$

Q059 **Written**

Simplify

$$\frac{x-1}{x^2+4x+3}+\frac{2}{x^2+5x+6}.$$

Classified Practice **Rational Expressions**

Q060 MCQ

Simplify

$$\frac{x-1}{(x+1)(x+3)}+\frac{2x+1}{(x+3)(x+5)}:$$

Choose the correct option.

A $\frac{x^3+12x^2+30x+11}{(x+1)(x+3)(x+5)}$

B $\frac{3x^2+7x-4}{(x+1)(x+3)(x+5)}$

C $-\frac{x^3+6x^2+16x+19}{(x+1)(x+3)(x+5)}$

D $-\frac{3x^2+7x-4}{(x+1)(x+3)(x+5)}$

Classified Practice **Rational Expressions**

Q061 MCQ

Simplify

$$\frac{x+3}{(x+2)(x+4)}+\frac{x-2}{(x+4)(x+6)}:$$

Choose the correct option.

A $\dfrac{x^3+14x^2+53x+62}{(x+2)(x+4)(x+6)}$

B $-\dfrac{x^3+10x^2+35x+34}{(x+2)(x+4)(x+6)}$

C $\dfrac{2x^2+9x+14}{(x+2)(x+4)(x+6)}$

D $-\dfrac{2x^2+9x+14}{(x+2)(x+4)(x+6)}$

Classified Practice **Rational Expressions**

Q062 MCQ

Simplify

$$\frac{3x+2}{(x+1)(x+4)}+\frac{x+5}{(x+4)(x+7)}:$$

Choose the correct option.

A $\dfrac{x^3+16x^2+68x+47}{(x+1)(x+4)(x+7)}$

B $-\dfrac{x^3+8x^2+10x+9}{(x+1)(x+4)(x+7)}$

C $-\dfrac{4x^2+29x+19}{(x+1)(x+4)(x+7)}$

D $\dfrac{4x^2+29x+19}{(x+1)(x+4)(x+7)}$

Classified Practice **Rational Expressions**

Q063 MCQ

Simplify

$$\frac{x+1}{(x+2)(x+5)}+\frac{4x-3}{(x+5)(x+8)}:$$

Choose the correct option.

A $\frac{5x^2+14x+2}{(x+2)(x+5)(x+8)}$

B $\frac{x^3+20x^2+80x+82}{(x+2)(x+5)(x+8)}$

C $-\frac{x^3+10x^2+52x+78}{(x+2)(x+5)(x+8)}$

D $-\frac{5x^2+14x+2}{(x+2)(x+5)(x+8)}$

Classified Practice

Rational Expressions

Q064 MCQ

Simplify

$$\frac{2x+1}{(x+3)(x+6)}+\frac{x+4}{(x+6)(x+9)}:$$

Choose the correct option.

A $\frac{x^3+21x^2+125x+183}{(x+3)(x+6)(x+9)}$

B $\frac{3x^2+26x+21}{(x+3)(x+6)(x+9)}$

C $-\frac{x^3+15x^2+73x+141}{(x+3)(x+6)(x+9)}$

D $-\frac{3x^2+26x+21}{(x+3)(x+6)(x+9)}$

Classified Practice **Rational Expressions**

Q065 MCQ

Simplify

$$\frac{x-2}{(x+1)(x+5)}+\frac{3x+1}{(x+5)(x+8)}:$$

Choose the correct option.

A $\frac{x^3+18x^2+63x+25}{(x+1)(x+5)(x+8)}$

B $-\frac{x^3+10x^2+43x+55}{(x+1)(x+5)(x+8)}$

C $\frac{4x^2+10x-15}{(x+1)(x+5)(x+8)}$

D $-\frac{4x^2+10x-15}{(x+1)(x+5)(x+8)}$

Classified Practice **Rational Expressions**

Q066 **Written**

Simplify

$$\frac{x}{x-1} - \frac{2}{x^2-1}.$$

Q067 **Written**

Simplify

$$\frac{x+2}{x}-\frac{x-2}{x-1}.$$

Classified Practice **Rational Expressions**

Q068 **Written**

Simplify

$$\frac{2}{x-1}+\frac{3}{x^2+5x-6}.$$

Classified Practice **Rational Expressions**

Q069 MCQ

Simplify
$$\frac{x+2}{(x+1)(x+3)} - \frac{2}{x+1}:$$

Choose the correct option.

A $\frac{x^2+3x-1}{(x+1)(x+3)}$

B $-\frac{x^2+5x+7}{(x+1)(x+3)}$

C $-\frac{x+4}{(x+1)(x+3)}$

D $\frac{x+4}{(x+1)(x+3)}$

Classified Practice

Rational Expressions

Q070 **Written**

Simplify

$$\frac{1}{x+2}+\frac{1}{x-2}.$$

Q071 **Written**

Simplify

$$\frac{x+4}{x^2+2x}-\frac{x+5}{x^2+x-2}.$$

Classified Practice

Rational Expressions

Q072 **Written**

Simplify

$$\frac{x+8}{x^2+x-2}-\frac{x+5}{x^2-1}.$$

Q073 **Written**

Simplify

$$\frac{x-3}{x^2+x}+\frac{x}{x^2-1}.$$

Classified Practice **Rational Expressions**

Q074 **Written**

Simplify

$$\frac{4x}{x^2-1}-\frac{x-1}{x^2+x}.$$

Q075 **Written**

Simplify

$$\frac{1}{x^2+5x+6}+\frac{1}{x^2+7x+12}.$$

Classified Practice **Rational Expressions**

Q076 Written

Simplify

$$\frac{x-1}{x^2+3x+2} - \frac{x-3}{x^2+4x+3}.$$

Classified Practice **Rational Expressions**

Q077 MCQ

A learner combines $\dfrac{2x-1}{(x-1)(x+2)} - \dfrac{x+4}{x+2}$. Correct the numerator line, then simplify the expression:

Choose the correct option.

A $\dfrac{1}{(x-1)(x+2)}$

B $-\dfrac{x^2+x-3}{(x-1)(x+2)}$

C $-\dfrac{2x^2+2x-5}{(x-1)(x+2)}$

D $\dfrac{x^2+x-3}{(x-1)(x+2)}$

Classified Practice **Rational Expressions**

Q078 **Written**

Simplify

$$\frac{x^3+8x}{x^2-x-2}-\frac{8}{x-2}.$$

Classified Practice **Rational Expressions**

Q079 **Challenge**

Simplify the single-inner fraction together with the reciprocal-pair quotient; state the full domain.

Classified Practice **Rational Expressions**

Q080 **Written**

Simplify

$$\frac{1}{1-\frac{1}{x-1}}.$$

Q081 **Written**

Simplify

$$\frac{1}{1-\dfrac{x}{x+1}}.$$

Classified Practice **Rational Expressions**

Q082 MCQ

Simplify

$$\frac{1}{1-\frac{x+2}{x+1}}:$$

Choose the correct option.

A $-x$

B $-x-2$

C $x+1$

D $-x-1$

Q083 **MCQ**

Simplify

$$\frac{1}{1-\frac{x+3}{x+2}}:$$

Choose the correct option.

A $-x-2$

B $-x-1$

C $-x-3$

D $x+2$

Classified Practice **Rational Expressions**

Q084 MCQ

Simplify

$$\frac{1}{1-\frac{2x+1}{x+1}}:$$

Choose the correct option.

A $-\frac{1}{x}$

B $-\frac{x+1}{x}$

C $-\frac{2x+1}{x}$

D $\frac{x+1}{x}$

Q085 **MCQ**

Simplify

$$\frac{1}{1-\frac{2x+3}{x+1}}:$$

Choose the correct option.

A $\frac{1}{x+2}$

B $-\frac{2x+3}{x+2}$

C $-\frac{x+1}{x+2}$

D $\frac{x+1}{x+2}$

Classified Practice **Rational Expressions**

Q086 MCQ

Simplify

$$\frac{1}{1-\frac{3x+1}{x+2}}:$$

Choose the correct option.

A $\frac{x-3}{2x-1}$

B $-\frac{3x+1}{2x-1}$

C $\frac{x+2}{2x-1}$

D $-\frac{x+2}{2x-1}$

Q087 **MCQ**

Simplify

$$\frac{1}{1-\frac{x+4}{x+3}}:$$

Choose the correct option.

A $-x-3$

B $-x-2$

C $-x-4$

D $x+3$

Classified Practice **Rational Expressions**

Q088 MCQ

Simplify

$$\frac{1}{1-\frac{2x+5}{x+1}}:$$

Choose the correct option.

A $\frac{3}{x+4}$

B $-\frac{x+1}{x+4}$

C $-\frac{2x+5}{x+4}$

D $\frac{x+1}{x+4}$

Q089 **Written**

Simplify

$$\frac{1}{1-x}+\frac{1}{1+x}\Big/\left(\frac{1}{1-x}-\frac{1}{1+x}\right).$$

Classified Practice

Rational Expressions

Q090 MCQ

Simplify

$$\frac{1}{x+1}+\frac{1}{x+4}\Big/\left(\frac{1}{x+1}-\frac{1}{x+4}\right):$$

Choose the correct option.

A $\frac{2x+5}{3}$

B $\frac{2(x+4)}{3}$

C $\frac{2(x+1)}{3}$

D $-\frac{2x+5}{3}$

Q091 **MCQ**

Simplify

$$\frac{1}{x+1}+\frac{1}{x+5}\Big/\left(\frac{1}{x+1}-\frac{1}{x+5}\right):$$

Choose the correct option.

A $\frac{x+3}{2}$

B $\frac{x+5}{2}$

C $\frac{x+1}{2}$

D $-\frac{x+3}{2}$

Classified Practice **Rational Expressions**

Q092 **Written**

Simplify

$$\frac{1}{x+2}+\frac{1}{x-2}\Big/\left(\frac{1}{x+2}-\frac{1}{x-2}\right).$$

Q093 MCQ

Simplify

$$\frac{1}{x+1}+\frac{1}{x+3}\Big/\left(\frac{1}{x+1}+\frac{1}{x+3}\right):$$

Choose the correct option.

A 2

B 0

C 1

D -1

Classified Practice **Rational Expressions**

Q094 MCQ

Simplify

$$\frac{1}{x+2} + \frac{1}{x+5} \Big/ \left(\frac{1}{x+2} + \frac{1}{x+5}\right):$$

Choose the correct option.

A 2

B 0

C -1

D 1

Q095 **MCQ**

Simplify

$$\frac{1}{x+3} + \frac{1}{x+6} \bigg/ \left(\frac{1}{x+3} + \frac{1}{x+6}\right):$$

Choose the correct option.

A 2

B 1

C 0

D -1

Classified Practice **Rational Expressions**

Q096 **Written**

Simplify

$$\frac{1}{x+y}+\frac{1}{x-y}\Big/\left(\frac{1}{x+y}-\frac{1}{x-y}\right).$$

Q097 **Written**

Simplify

$$\frac{1}{1-\frac{1}{1-\frac{1}{x}}}.$$

Classified Practice **Rational Expressions**

Q098 **Written**

Simplify

$$\frac{1}{1-\frac{1}{1+\frac{1}{x}}}.$$

Q099 **Written**

Simplify

$$\left(1+\frac{x-y}{x+y}\right)\Big/\left(1-\frac{x-y}{x+y}\right).$$

Classified Practice **Rational Expressions**

Q100 **Written**

Simplify

$$\left(x-\frac{2}{x-1}\right)\Big/\left(\frac{x}{x-1}-2\right).$$

Q101 **Written**

Simplify

$$\left(\frac{x+1}{x-1}-\frac{x-1}{x+1}\right)\Big/\left(\frac{x+1}{x-1}+\frac{x-1}{x+1}\right).$$

Classified Practice **Rational Expressions**

Q102 **Written**

Simplify

$$\frac{1}{1+\frac{1}{1+\frac{1}{1+x}}}.$$

Quiz MCQ 1 **Concept Quiz · MCQ**

Simplify $\frac{x^2-4}{3x^2+7x+2}$:

Choose the correct option.

A $\frac{x-2}{3x+1}$

B $\frac{x+2}{3x+1}$

C $\frac{x-2}{3x+2}$

D $\frac{x^2-4}{3x+1}$

Classified Practice **Rational Expressions**

Quiz MCQ 2 **Concept Quiz · MCQ**

Simplify

$\dfrac{x+1}{x-2} + \dfrac{2}{x-2}$:

Choose the correct option.

A $\dfrac{x+3}{x-2}$

B $\dfrac{x-1}{x-2}$

C $\dfrac{x+1}{2x-4}$

D $\dfrac{x+3}{x+2}$

Quiz MCQ 3 **Concept Quiz · MCQ**

In $\frac{3}{x^2+x} - \frac{3-x}{x^2+x}$, the numerator after the minus sign is... Which of the following is correct?:

Choose the correct option.

A $3-(3-x)$

B $3-3-x$

C $3+3-x$

D $3-x$

Classified Practice **Rational Expressions**

Quiz MCQ 4 **Concept Quiz · MCQ**

Which statement is safe when simplifying a compound fraction?:

Choose the correct option.

A Multiply top and bottom by the same inner denominator
B Add all denominators
C Cancel terms across +
D Drop all restrictions

Quiz WRITTEN 5 Concept Quiz · Written

Simplify $\frac{x^2-9}{x^2-2x-3} \times \frac{x^2-4}{x^2+x-6}$ and state the original restrictions.

Classified Practice **Rational Expressions**

Quiz WRITTEN 6 Concept Quiz · Written

Simplify
$\dfrac{x^2-4}{x+1}+\dfrac{3}{x+1}$ **and state the original restriction.**

Quiz WRITTEN 7 **Concept Quiz · Written**

Simplify

$\frac{x^2 - y^2}{x^2 - 2xy + y^2} - \frac{x + y}{x - y}$ **in x and y.**

Classified Practice **Rational Expressions**

Quiz WRITTEN 8 **Concept Quiz · Written**

Explain why $(x^2-16)/(x-4)=x+4$ is not an identity for every real x.

Answer key

Use the question number shown on each page.

Q001 A

Q002 B

Q003 C

Q004 D

Q005 A

Q006 B

Q007 $\frac{4b^3}{3a^2}$

Q008 $\frac{3y^2}{10x}$

Q009 $\frac{3a}{4b^3}$

Q010 $\frac{1}{4ab^2}$

Q011 B

Q012 A

Q013 A

Q014 B

Q015 A

Q016 B

Q017 C

Q018 D

Q019 A

Q020 B

Q021 C

Q022 $\frac{3c^2}{a}$

Q023 $\frac{4a^2c^3}{b^2}$

Q024 C

Q025 D

Q026 D

Q027 A

Q028 B

Q029 C

Q030 $\frac{x-2}{3x+1}$

Q031 $\frac{2x-5}{x+2}$

Q032 $\frac{x}{x-3}$

Q033 $\frac{x+2}{x-4}$

Q034 $\frac{2(x-y)}{x(x+4y)}$

Q035 $\frac{x^2+x+1}{x+3}$

Q036 $\frac{x}{x-3}$

Q037 $\frac{x-4}{x(x-2)}$

Q038 $\frac{(x-5)(x+3)}{x(x+4)}$

Q039 $\frac{(x+3)(x+4)}{(2x-3)(3x+1)}$

Q040 $\frac{(x-3)(x+1)}{(x+2)(x+4)}$

Q041 $\frac{x-3}{(x-2)(2x-1)}$

Q042 $\frac{3ab^2}{4x^2}$

Q043 $\frac{(x-3)(2x-5)}{(x-1)(x+1)}$

Q044 $\frac{3}{x+3}$

Q045 $\frac{1}{x}$

Q046 $-\frac{1}{2}+\frac{15129\sqrt{123}}{2}$

Q047 $\frac{x^2+4x-2}{(x-2)(x+3)}$

Q048 $x-1$

Q049 C

Q050 D

Q051 A

Q052 B

Q053 $\frac{1}{x+1}$

Q054 A

Q055 B

Q056 C

Q057 C

Q058 $\frac{x+2}{x(x-1)}$

Q059 $\frac{x}{(x+1)(x+2)}$

Q060 B

Q061 C

Q062 D

Q063 A

Q064 B

Q065 C

Q066 $\frac{x+2}{x+1}$

Q067 $\frac{3x-2}{x(x-1)}$

Q068 $\frac{2x+15}{(x-1)(x+6)}$

Q069 C

Q070 $\frac{2x}{(x-2)(x+2)}$

Q071 $-\frac{2}{x(x-1)}$

Q072 $\frac{2}{(x+1)(x+2)}$

Q073 $\frac{2x^2-4x+3}{x(x-1)(x+1)}$

Q074 $\frac{3x-1}{x(x-1)}$

Q075 $\frac{2}{(x+2)(x+4)}$

Q076 $\frac{3}{(x+2)(x+3)}$

Q077 B

Q078 $\frac{x^2+2x+4}{x+1}$

Q079 $-\frac{x^2-4x+2}{2(x-2)}$

Q080 $\frac{x-1}{x-2}$

Q081 $x+1$

Q082 D

Q083 A

Q084 B

Q085 C

Q086 D

Q087 A

Q088 B

Q089 $1/x$

Q090 A

Q091 A

Q092 $-\frac{x}{2}$

Q093 C

Q094 D

Q095 B

Q096 $-\frac{x}{y}$

Q097 $1-x$

Q098 $x+1$

Q099 $\frac{x}{y}$

Q100 $-(x+1)$

Q101 $\frac{2x}{x^2+1}$

Q102 $\frac{x+2}{2x+3}$

Quiz MCQ 1 A

Quiz MCQ 2 A

Quiz MCQ 3 A

Quiz MCQ 4 A

Quiz WRITTEN 5 $\frac{x+2}{x+1}$

Quiz WRITTEN 6 $x-1$

Quiz WRITTEN 7 0

Quiz WRITTEN 8 $x \neq 4;$ removable hole at $(4,8)$